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Info | How it works | Prize | Partners | FAQ

IT TAKES ONE IDEA TO MAKE AN IMPACT

Red Bull partners with Microsoft & AMD to give you the AI tools, mentorship and a global platform to bring your idea to life – and the wiings to pitch it in the Silicon Valley.

Global prize

$100K CASH

to bring your idea to life

HOW IT WORKS

01 Application
Brainstorm or refine an idea with the AI Mentors and submit it.

02 National final
The best proofs of concept will be eligible to win.

03 Development phase
National winners will get access to exclusive AI tools & mentoring.

04 World final
3-day experience in Silicon Valley with global exposure and network.

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Presentation Outline

- Introduction
- Definition of Terms
- Background to the Research
- Problem Identification
- Practical Approaches to Problem Solving
- The Minimum Viable Product (MVP)
- Challenging Assumptions in Innovation and Practical Approaches
- Understanding User Needs and Practical Approaches
- Structured Frameworks in the Innovation Lifecycle
- Refining Ideas
- Practical Innovation Workflow
- Conclusion
- References

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Definitions Cont'd

- **Problem solving.** Systematic, data-driven technique used to bridge the gap between an uncertain current state and a desired objective [ASQ, 2025]. In the context of 4IR, it is increasingly defined through computational thinking (decomposing complex systems into manageable algorithmic components and abstractions) [Tang et al., 2024]. Modern frameworks, such as the RISE Model (2025), emphasize a holistic approach: Recognizing the core issue, Investigating through real-time data, Strategizing with creative "Growth Mindsets," and Executing with agility [Daily Focus, 2025; Aurora Training, 2025].

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Definitions Cont'd

- **Innovation.** Process of translating an idea or invention into a product, service, or process that creates value or for which customers will pay [Inso, 2025] defined by the "fusion of technologies blurring the lines between the physical, digital, and biological spheres" [Schwab, 2016].
- **Invention.** The initial spark of creation—the first time a completely new product, process, or system is brought into existence [Schilling, 2018]. Representing a significant technical or scientific breakthrough, it is often a "raw" discovery that exists in a lab or as a patent before it has been refined to solve a practical problem for the general public

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Problem Identification

- To identify real-world problems that demand innovative solutions, one must move beyond surface-level symptoms to find gaps that are pervasive, urgent, and expensive [ASQ, 2025].
- In the 4IR, this process is increasingly human-centric, requiring the strategy of gaining direct, firsthand experience by being physically being present where the work is happening or where the problem exists.
- Instead of relying on second-hand reports, data dashboards, or theoretical models from an office, an innovator engages in direct observation and interacts with the people facing the challenge [Amper, 2025; ProjectManager, 2025].

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Practical Approaches cont'd

4. Challenging Assumptions through Pre-Mortems

Before committing to a solution, innovators identify problems by imagining a future failure. By conducting a Pre-Mortem (a strategic thinking exercise where a team imagines that a project has already failed and then works backward to determine what potentially caused that failure), teams work backward to find the "hidden" problems—such as lack of user trust or ethical biases in AI—that could kill an innovation before it starts [Harvard Business Review, 2024].

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Key Characteristics in Innovation

- **Challenging Assumptions:** It forces the team to voice "politically sensitive" or "unpleasant" risks that are often ignored during the honeymoon phase of an idea [Paul & Elder, 2019].
- **Identifying "Kill-Switch" Risks:** It helps innovators find the single points of failure that could stop an MVP from scaling [Ries, 2011].
- **Psychological Safety:** By making "failure" the starting point of the conversation, it makes it safe for team members to be critical without appearing "negative" or "unsupportive" [Harvard Business Review, 2024].

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The Minimum Viable Product (MVP)

- A Minimum Viable Product (MVP) is the simplest version of a new product that allows a team to collect the maximum amount of validated learning about customers with the least effort [Ries, 2011]. It is not a "lite" or "broken" version of a product; it is a functional tool designed specifically to test a "leap-of-faith" assumption [Strategyzer, 2025].
- **Key Characteristics in Innovation:**
  - **The "Minimum" Scope:** It focuses only on the core value proposition. If removing a feature doesn't stop the product from solving the primary problem, it is excluded from the MVP [Itonics, 2025].
  - **Validated Learning:** The goal is to move through the Build-Measure-Learn feedback loop as quickly as possible to decide whether to pivot or persevere [Ries, 2011].
  - **Customer Feedback:** It is used to observe how real users interact with a solution, rather than relying on what they *say*

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Challenging Assumptions in Innovations

- Challenging assumptions is the "immune system" of innovation; it prevents a team from investing in a solution that is technically possible but fundamentally flawed in its logic. In the **Fourth Industrial Revolution (4IR)**, where technologies like AI are often over-hyped, critical thinking is required to separate "leap-of-faith" facts from dangerous fantasies [Ries, 2011; Harvard Business Review, 2024].

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Practical Approaches to Challenging Assumptions

1. Assumption Mapping: Identifying the "Kill-Switch"

- Not all assumptions are equal. Innovators use a 2x2 matrix to plot assumptions based on Importance (how vital they are to success) and Certainty (how much evidence already exists). The goal is to focus exclusively on the High Importance / Low Certainty quadrant—these are the assumptions that could "kill" the project if they are wrong [Strategyzer, 2025].

Scenario: Assuming that "users will prefer biometric login over passwords" is a high-risk assumption that must be tested before building the hardware.

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Practical Approaches cont'd

3. Applying "First Principles" Thinking

- Popularized by innovators like Elon Musk, this involves breaking a problem down to its fundamental physical or logical truths and rebuilding a solution from scratch, rather than relying on "the way it has always been done" [Schilling, 2018].

Scenario: Instead of assuming "we need a faster database," First Principles thinking asks, "Why do we need to store this data at all?" This challenges the core assumption of the product's architecture.

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Practical Approaches cont'd

4. The "Mom Test": Seeking Truth Over Approval

When interviewing users, innovators often accidentally "fish" for compliments, leading to false validation. Challenging assumptions requires asking about past behaviors and specific facts rather than hypothetical opinions about a future product [Fitzpatrick, 2024].

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Understanding User Needs

To understand user needs effectively, innovators must move beyond asking customers what they want and instead focus on uncovering unarticulated or latent needs. In the Fourth Industrial Revolution, where digital interactions often mask human friction, identifying these needs requires a combination of observational empathy and rigorous behavioral data [Nature Scientific Reports, 2025; Schwab, 2016]

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Practical Approaches to Understanding User Needs

1. The "Jobs-to-be-Done" (JTBD) Framework

People do not buy products; they "hire" them to settle a specific struggle or achieve a progress-oriented goal. Understanding user needs involves identifying the functional, emotional, and social "job" a user is trying to accomplish [Strategyn, 2025].

Security Example: A user doesn't want "biometric encryption"; they want the "peace of mind" that their personal data is safe without the "hassle" of remembering passwords.

2. Empathy Mapping and "Deep Inquiries"

Critical thinking requires looking past stated needs (what users say) to find latent needs (what they actually do). This is achieved through "Deep Inquiries"—interviews that focus on past behaviors and specific pain points rather than hypothetical future desires [Fitzpatrick, 2024; Paul & Elder, 2019].

Practical Tool: Use the Mom Test to ask: *"Talk me through the last time you experienced a security breach,"* instead of *"Would you like an AI-driven security app?"* [Fitzpatrick, 2024].

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Practical Approaches to Understanding User Needs

3. Behavioral Observation (The "Gemba" Walk)

Directly observing users in their "natural habitat" reveals the "workarounds" and "hacks" they have created to solve problems that existing tools ignore. These physical or digital "pains" are the most reliable indicators of a real-world problem [Amper, 2025; ProjectManager, 2025].

Scenario: If you see employees writing passwords on sticky notes, the "need" isn't more security training; it is a frictionless identity management system.

4. Analyzing "Lead Users" and Extremes

Innovation often comes from studying "Lead Users"—those who are ahead of market trends and have already modified existing technologies to meet their sophisticated needs [Schilling, 2018]. Understanding the needs of these "extreme" users often predicts the future needs of the mainstream market.

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Structured Frameworks in the Innovation Lifecycle

1. The "5 Whys" and Root Cause Analysis (RCA)

- Before innovating a solution, teams must ensure they are solving the right problem. The 5 Whys technique involves repeatedly asking "Why?" to peel away surface symptoms and reveal the systemic failure [ASQ, 2025].
  - Why are we vulnerable? *Hackers are using stolen session tokens.*
  - Why can they use stolen tokens? *Our system trusts any token until it expires.*
  - Why does it trust them? *We use 'Castle and Moat' security logic.*
  - Why haven't we changed it? *Legacy systems make micro-segmentation difficult.*
  - Root Cause: We lack a Continuous, AI-Driven Identity Verification system that adapts to user behavior [Lean.org; ASQ, n.d.].

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Structured Frameworks in the Innovation Lifecycle

2. The RISE Model (Recognize, Investigate, Strategize, Execute)

- Modern corporate environments utilize the RISE framework (2025) to manage the non-linear nature of innovation.
- Recognize & Investigate: Using Gemba Walks to identify inefficiencies [ProjectManager, 2025].
- Strategize & Execute: Leveraging Computational Thinking to design algorithmic solutions that can scale within 4IR digital ecosystems [Tang et al., 2024; Daily Focus, 2025].

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Structured Frameworks in the Innovation Lifecycle

3. Lean Startup: Build-Measure-Learn

- This framework treats innovation as a series of experiments. Instead of building a final product, teams develop a Minimum Viable Product (MVP) to test "leap-of-faith" assumptions [Ries, 2011].
- Innovation Context: This structured loop ensures that data-informed decisions—rather than "vanity metrics"—drive the refinement of the solution [Ries, 2011; Strategyzer, 2025].

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Refining Ideas Through Rapid Prototyping and Data-Informed Decision Making

- In the innovation lifecycle, refining an idea is the process of moving from a high-risk concept to a validated, scalable solution. In the context of the 4IR, refining IT solutions requires a transition from "static" development to dynamic, data-driven evolution. The focus is on reducing the "Cost of Failure" through rapid prototyping and the transition from gut-feeling to data-informed decision-making [Ries, 2011; Strategyzer, 2025].

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Refining Ideas cont'd

1. Rapid Prototyping: Thinking with Your Hands

- Rapid prototyping is a physical way of asking a question [Itonics, 2025]. In 4IR, IT innovators use "low-fidelity" tools to test core logic before writing expensive code.
- **The MVP as a Learning Tool**: MVP is designed to test a specific "leap-of-faith" assumption [Ries, 2011]. For example, a "Shadow Mode" MVP in AI security might run an algorithm in the background to see if it correctly identifies threats without actually blocking traffic, thus testing technical accuracy without risking system downtime [Itonics, 2025; Cloud4C, 2026].
- Types of Prototypes: Innovators use "Concierge" MVPs (manually performing the AI's job) or "Wizard of Oz" MVPs (the front-end looks automated, but a human is behind it) to validate user needs before full-scale automation [Itonics, 2025; Ries, 2011].

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Refining Ideas cont'd

2. Data-Informed Decision Making

- Refining an innovation requires a shift from "Vanity Metrics" (data that makes you look good but doesn't track progress, like total downloads) to "Actionable Metrics" (data that links specific actions to outcomes) [Ries, 2011; ASQ, 2025].
- **The Build-Measure-Learn Loop:** Data from each prototype informs whether to Persevere (keep going), Pivot (change the strategy), or Perish (stop the project) [Ries, 2011; Daily Focus, 2025].
- **Computational Thinking:** In 4IR, decision-making is increasingly augmented by AI. Innovators use real-time data analytics and digital twins to simulate how a security solution will perform across a global network, providing "synthetic" data for refinement when real-world data is scarce [Nature Scientific Reports, 2025; Tang et al., 2024].

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Refining Ideas cont'd

3. Overcoming Cognitive Bias in Refinement

- Critical thinking is essential during refinement to combat "Sunk Cost Fallacy"—the tendency to keep investing in a failing prototype because of the time already spent. By using Assumption Mapping, teams ensure they are only iterating on features that are vital to the project's success [Strategyzer, 2025; Harvard Business Review, 2024].

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Practical Innovation Workflow

AI-Driven Zero Trust Security

Phase 1: Critical Problem Identification (The "Inquiry" Stage)

- **The Problem:** "Our cloud databases are vulnerable to unauthorized access."
- **Applying Critical Thinking:** Challenging the "Stated Need." Is the problem actually *access*, or is it *static identity verification*?
- **Metacognitive Check:** Avoid Availability Bias (assuming the solution is just "better passwords" because that's what we know).
- **The Refined Problem:** "In a 4IR environment, static credentials (passwords/MFA) are no longer sufficient to verify identity in real-time." [Paul & Elder, 2019; Kahneman, 2011].

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Practical Innovation Workflow

Phase 3: Idea Validation & MVP Scoping (The "Scientific" Stage)

- **Innovation Idea:** Deploy an Agentic AI that monitors "patterns of life" (typing speed, IP geolocation, access times) to continuously re-verify users.
- **Critical Thinking (Assumption Mapping):** Identify the "Kill-Switch" assumption: *Can an AI distinguish between a user and a sophisticated bot without lagging the system?*
- **MVP Scope:** Instead of a full rollout, build a "Shadow Mode" Prototype. The AI monitors traffic and flags anomalies but doesn't block anyone yet. This gathers Actionable Data without breaking the live IT environment [Ries, 2011; Strategyzer, 2025].

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Practical Innovation Workflow

Phase 4: Solution Refinement (The "Iterative" Stage)

- **Data-Informed Decision Making:** Analyze the "False Positive" rate from the Shadow Mode test.
- **Systems Thinking:** Evaluate how this new AI security layer interacts with the rest of the IT stack. Does it slow down the DevOps pipeline? [The Systems Thinker, n.d.].
- **Final Solution:** Refine the AI to use **Retrieval-Augmented Generation (RAG)** to pull real-time threat intelligence from external databases, ensuring it recognizes new "Zero-Day" attack patterns instantly [IBM, 2024].

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Conclusion

- In her book "***Quirky***", Melissa Schilling argues that breakthrough innovation is driven by a distinct psychological temperament rather than just raw intelligence [Schilling, 2018]. By analyzing icons like Nikola Tesla and Marie Curie, she identifies key traits: high self-efficacy, social detachment, and extreme perseverance. These "quirks" allow innovators to maintain an unshakable belief in their vision, ignore social critics, and endure years of repeated failure that would break others.
- Schilling highlights that these geniuses often possess multiple cognitive perspectives, pulling analogies from diverse fields to create novel, "T-shaped" solutions. They are fueled by a grand sense of purpose and idealism that transcends financial gain. Ultimately, the book serves as a challenge to modern corporate culture: to foster true disruption, organizations must move beyond "team-player" norms and provide the psychological safety needed for eccentric, non-conforming minds to flourish and solve the world's most complex problems.

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